Disease and Remedy Information Management System

Project Plan: frontend first, backend second

How the parts are marked

Tag	Meaning	When it is built
[FRONTEND]	Everything the visitor or admin sees and clicks: pages, design, components, interface checks.	Phase 1 (first)
[BACKEND]	Server, database, login, security and endpoints. Planned now, built later.	Phase 2 (after the frontend)
[LINK]	The bridge: the shared data structure and the services area. Built on the frontend side but shaped by the backend.	Defined in Phase 1, completed in Phase 2
[CONTENT]	The health information itself: sample records, sources and content rules.	Throughout
[BOTH]	Items that touch the whole system.	Throughout

Working site name: **DiseaseWise** (can be changed at any time). Prepared October 2026. This document covers planning only; it contains no code.

Contents

Part 1: Project overview [BOTH] 4

1.1 Purpose . . . 4
1.2 Problems it solves . . . 4
1.3 Users . . . 4
1.4 Major features . . . 4
1.5 Build approach . . . 4
1.6 Decisions made . . . 4

Part 2: Frontend plan [FRONTEND] 6

2.1 Pages . . . 6
2.2 Wireframes in words . . . 6
2.3 Components . . . 7
2.4 Technology and organization . . . 7
2.5 Visual design . . . 8
2.6 Branding . . . 9

Part 3: Content plan [CONTENT] 10

3.1 Starter diseases . . . 10
3.2 Fields prepared for every disease . . . 10
3.3 Content rules . . . 10
3.4 Sample record: Malaria . . . 10
3.5 Edge cases to include on purpose . . . 11

Part 4: The link between frontend and backend [LINK] 12

4.1 Data structure . . . 12
4.2 The services area . . . 12
4.3 Search rules (mirrored by the backend) . . . 12
4.4 Endpoints the frontend will eventually call . . . 12
4.5 Response rules . . . 13

Part 5: Backend plan [BACKEND] 14

5.1 Goals . . . 14
5.2 Technology . . . 14
5.3 How the system fits together . . . 14
5.4 Database design . . . 14
5.5 Search design . . . 15
5.6 Authentication and security . . . 15
5.7 Content integrity rules . . . 15

5.8 Hosting needs . . . 15

Part 6: Frontend build plan [FRONTEND] 16

Milestone 1: Setup and design foundations (3 to 4 sessions) . . . 16

Milestone 2: Shared layout and data foundation (3 to 4 sessions) . . . 16

Milestone 3: Home page and search (4 to 5 sessions) . . . 16

Milestone 4: Disease and remedy pages (4 to 5 sessions) . . . 16

Milestone 5: Category and About pages (2 to 3 sessions) . . . 17

Milestone 6: Admin area (6 to 8 sessions) . . . 17

Milestone 7: Responsive, accessibility and polish (3 to 4 sessions) . . . 17

Milestone 8: Testing and handover (3 to 4 sessions) . . . 17

Part 7: Backend build plan [BACKEND] 19

B1: Setup, database and sample data (3 to 4 sessions) . . . 19

B2: Public endpoints (4 to 5 sessions) . . . 19

B3: Connect the public frontend pages (3 to 4 sessions) . . . 19

B4: Admin login and protection (3 to 4 sessions) . . . 19

B5: Admin endpoints (5 to 6 sessions) . . . 19

B6: Connect the admin frontend pages (3 to 4 sessions) . . . 20

B7: Security review and full testing (3 to 4 sessions) . . . 20

B8: Deployment (2 to 3 sessions) . . . 20

Part 8: Testing checklists 21

8.1 Frontend checklist [FRONTEND] . . . 21

8.2 Content checks [CONTENT] . . . 21

8.3 Readiness for the backend [LINK] . . . 21

8.4 Backend checklist [BACKEND] . . . 22

Part 9: Timeline, risks and open decisions 23

9.1 Timeline (about 4 sessions a week, 2 to 3 hours each) . . . 23

9.2 Risks . . . 23

9.3 Open decisions . . . 24

9.4 Version 2 ideas (not in the first build) . . . 24

Part 1: Project overview [BOTH]

1.1 Purpose

A website where people search for a disease and read about its symptoms, causes, risk factors, prevention and medically recognized treatment information. An administrator manages the content. The site is informational only: it educates users and does **not** diagnose illnesses or replace a doctor.

1.2 Problems it solves

- Organized disease information is hard to access, and it is scattered across websites and books.
- Users struggle to find prevention and treatment information in one place.
- An administrator needs a simple way to add, update and manage disease information.

1.3 Users

User	What they do
Public visitor	Searches and reads. No login needed.
Administrator	Logs in and adds, edits and deletes diseases, remedies and categories.

1.4 Major features

Feature	Description	Part
Home page	Introduces the site and its purpose, with a search box for diseases.	**[FRONTEND]**
Disease information page	Disease name, description, symptoms, causes, risk factors and prevention.	**[FRONTEND]**
Remedy information page	Reviewed treatment information, care options, precautions, and advice on when to seek professional care.	**[FRONTEND]**
Administrator dashboard	Authorized admin adds, edits and deletes disease and remedy records and organizes them into categories.	**[FRONTEND]** screens, **[BACKEND]** enforcement
Data storage, search and login	Permanent storage, search, and protection of admin features.	**[BACKEND]**

1.5 Build approach

- **[FRONTEND]** is built first, using sample data shaped exactly like the future backend responses.
- **[LINK]** Every page gets data only through one services area. Later, only that area changes to call the real backend.
- **[BACKEND]** is planned in full now (database, endpoints, security) but built only after the frontend is tested and handed over.

1.6 Decisions made

Decision	Choice	Part
Frontend technology	React with Vite, React Router and Tailwind CSS	**[FRONTEND]**
Backend technology	Node.js with Express	**[BACKEND]**

Decision	Choice	Part
Database	MySQL	**[BACKEND]**
Password protection	bcrypt hashing	**[BACKEND]**
Design mood	Calm teal palette, mobile first	**[FRONTEND]**
Working name and logo	DiseaseWise, magnifier-over-cross icon	**[FRONTEND]**

Part 2: Frontend plan [FRONTEND]

2.1 Pages

Public pages

Page	Route	What it shows
Home	/	Purpose, search box, popular diseases, browse by category, how it works, disclaimer
Search results	/search?q=...	Matching diseases with summaries, category filter, no-results state
Disease	/disease/:name	Name, description, symptoms, causes, risk factors, prevention, link to remedy
Remedy	/disease/:name/remedy	Treatment overview, care options, precautions, when to seek professional care, source
All categories	/categories	Grid of categories with disease counts
Single category	/category/:name	Diseases in one category, alphabetical sort
About and disclaimer	/about	Purpose, limits, sourcing, consult a professional
Not found	any other address	Friendly message with a search bar and link home

Admin pages (screens only until the backend exists)

Page	Route	What it shows
Login	/admin/login	Email, password, error messages
Dashboard	/admin	Totals, recently updated, quick actions
Disease list	/admin/diseases	Searchable table with edit and delete
Add/edit disease	/admin/diseases/new and /:id/edit	Form with repeatable lists for symptoms, causes, risk factors, prevention
Remedy pages	/admin/remedies	List and form tied to one disease
Categories	/admin/categories	Add, rename, delete with in-use block

Until the backend exists, admin screens work on sample data only. Changes reset on refresh, and the login is a visual placeholder. **[BACKEND]** later enforces real login.

2.2 Wireframes in words

Shared on public pages

- Navbar: logo left, Home, Categories and About on the right, collapsing menu on phones.
- Disclaimer banner: slim, calm strip saying the site is educational and not a replacement for a doctor.
- Footer: purpose line, links and repeated disclaimer.

Home

- Hero with headline, one supporting line and a large search box with button.
- Popular diseases: 4 to 6 cards (name, one-line summary, category label).
- Browse by category: cards with icon and disease count.
- How it works: three steps (Search, Read, Know when to seek care).

- Disclaimer note and footer.

Search results

- Search bar pre-filled with the query; result count line.
- Category filter: side panel on desktop, collapsible on phones.
- Result cards: name, two-line summary, category tag, Read more.
- No-results state with spelling hints and a link to categories; loading placeholders.

Disease page

- Breadcrumbs, header block (name, category tag, description).
- Quick-jump menu (sticky on large screens).
- Sections in order: Description, Symptoms, Causes, Risk factors, Prevention.
- Treatment call-to-action card, urgent care box, disclaimer.

Remedy page

- Breadcrumbs, header with link back to the disease.
- Overview, recommended care options, precautions (amber box).
- When to seek professional care: the most prominent box on the page (red).
- Source and review date, disclaimer. A disease with no remedy shows a clear 'being prepared' message.

Admin layout

- Left sidebar (Dashboard, Diseases, Remedies, Categories, Logout) and a top bar with the admin name.
- On phones the sidebar becomes a slide-out menu and tables scroll sideways.

2.3 Components

Group	Components
Shared	Navbar, Footer, Disclaimer Banner, Button, Tag, Breadcrumbs, Loading Indicator, Empty State, Error Message
Search and browsing	Search Bar (with suggestions), Search Result Card, Category Card, Category Filter, Disease Card
Disease and remedy	Disease Header, Info Section (reused for symptoms, causes, risk factors, prevention), Quick Jump Menu, Treatment Call-to-Action, Warning Box (amber), Urgent Care Box (red), Source Note
Admin	Sidebar, Stat Tile, Data Table, Disease Form, Remedy Form, Repeatable List Input, Confirm Delete Popup, Form Field, Notice Bar

2.4 Technology and organization

Need	Choice
Interface building	React
Setup and dev tool	Vite
Moving between pages	React Router
Styling	Tailwind CSS
Sample data	Local data files (stand-in for the backend)
Forms (optional)	React Hook Form

Need	Choice
Icons (optional)	Lucide or React Icons

- Separate the public site from the admin site; they share components but not layouts.
- **[LINK]** All data requests go through one services area; pages never read sample data directly.
- One reusable section component serves symptoms, causes, risk factors and prevention.
- The frontend admin login is a convenience only; real protection comes from the **[BACKEND]**.
- The disclaimer is one shared component so wording is identical everywhere.

Project folder organization

Area	Purpose
Pages	One item per screen
Components	Reusable building blocks
Layouts	Public frame and admin frame
Routes	Web addresses and which page each opens
Data	Sample diseases, remedies, categories
Services	The single place where all data requests happen
Utilities	Small helpers (for example, name to web-friendly address, date formatting)
Assets	Logo, icons, images
Styles	Global styling and Tailwind setup

2.5 Visual design

Role	Color	Used for
Primary	Teal green #0F766E	Buttons, links, navbar accents, headings
Primary light	Soft mint #CCFBF1	Backgrounds, tags, hover states
Background	Off-white #F8FAFC	Page background
Text	Dark slate #1E293B	Main text
Secondary text	Gray-slate #64748B	Summaries, captions, dates
Warning	Amber #D97706	Precautions only
Urgent	Red #DC2626	'When to seek professional care' only
Success	Green #16A34A	Admin confirmations

Rule: red and amber are reserved for safety messages. Anywhere else they lose their meaning.

- Typography: clean sans-serif headings (for example Poppins or Inter semi-bold) and readable body text (Inter or Open Sans), 16 to 18px on phones with generous spacing.
- Layout: narrow reading column, generous white space, rounded cards, one column on phones, two on tablets, three or four on desktops.
- Search box is the most prominent element on the home page; the red box is the strongest block on the remedy page.
- Icons from one consistent line-icon set; minimal imagery and no graphic medical photos.

- Accessibility: strong contrast, never color alone for warnings, keyboard use with visible focus, labels on every form field, descriptive link text.

2.6 Branding

Item	Plan
Name	Working name DiseaseWise (alternatives: Remedia, HealthPedia, MedInfo Hub, HealthLens). Check no existing health site uses it and whether the web address is available.
Tagline	'Clear health information in one place.'
Logo	Magnifier over a cross, name in heading font, teal. Versions: full, icon-only, white. Export as SVG.
Appears in	Navbar, home hero, footer, browser tab, admin login and sidebar, share preview

Part 3: Content plan [CONTENT]

3.1 Starter diseases

Disease	Category	Why it is useful
Malaria	Infectious	The project's own example, very relevant locally
Typhoid fever	Infectious	Second infectious record, prevention-heavy
Hypertension	Chronic	Long-term condition with lifestyle-based care
Type 2 diabetes	Chronic	Many risk factors, tests long lists
Asthma	Respiratory	Third category, strong 'seek urgent care' section

3.2 Fields prepared for every disease

- Disease: name, web-friendly name, category, short summary, description, symptoms (5 to 8), causes (2 to 4), risk factors (3 to 6), prevention (4 to 6).
- Remedy: overview, care options, precautions, when to seek professional care, source or reviewer, last reviewed date.

3.3 Content rules

- No dosages or specific prescription instructions; name treatment types and point to a health professional.
- Plain language for an ordinary reader; similar list lengths across diseases.
- Every remedy cites a source. Reliable references: WHO, NCDC, CDC, NHS, Mayo Clinic. Rewrite in your own words and never copy text directly.
- Have a qualified person (pharmacist, nurse, doctor or lecturer) read the content before it is relied on.

3.4 Sample record: Malaria

Drafted in plain language from WHO material. Lines marked (verify) must be checked against NCDC guidance before publishing.

Field	Content
Name / slug / category	Malaria / malaria / Infectious diseases
Short summary	Malaria is a serious but preventable and treatable illness spread by mosquito bites. Early testing and treatment make a big difference.
Description	Malaria is a life-threatening disease caused by Plasmodium parasites, passed to people through the bites of infected female Anopheles mosquitoes. It is preventable and treatable, and early diagnosis and treatment reduce illness, prevent deaths and help reduce spread.
Symptoms	Fever, headache and chills (can be mild and easy to mistake for other illnesses); vomiting; symptoms usually start 10 to 15 days after the bite; may be milder in people who have had malaria before. Severe signs: extreme tiredness, impaired consciousness, repeated convulsions, difficulty breathing, dark or bloody urine, jaundice, abnormal bleeding.
Causes	Infection with Plasmodium parasites from an infected female Anopheles mosquito; less commonly, blood transfusions or contaminated needles.
Risk factors	Children under 5, pregnant people, people with HIV/AIDS, travelers with no previous immunity; living in or traveling to areas where malaria is common; sleeping without mosquito protection.

Field	Content
Prevention	Insecticide-treated nets, repellent, protective clothing; indoor residual spraying; the RTS,S vaccine for children in moderate to high transmission areas; preventive medicines for certain groups (for example pregnant women, young children, travelers) under health-worker guidance.
Remedy overview	Malaria always needs treatment with medicine. For the most common dangerous type, the best available treatment is artemisinin-based combination therapy (ACT). The right medicine depends on the type and the patient and must be chosen by a health professional.
Care options	Get tested early; take the treatment prescribed by a health professional; finish the full course (verify); rest and drink plenty of fluids (verify).
Precautions	Do not use leftover or unprescribed medicine (verify); do not assume every fever is malaria, testing confirms it; keep using mosquito protection during recovery.
When to seek care (red box)	Anyone with severe symptoms needs emergency care right away; seek prompt testing for fever in a child, pregnant person or someone with a weak immune system; seek care if fever starts after visiting a malaria area; seek care if symptoms do not improve after starting treatment.
Source and review	WHO malaria fact sheet and health topic pages; local check against NCDC or Federal Ministry of Health guidance; last reviewed: date of your own check.

3.5 Edge cases to include on purpose

- One disease with a very long symptoms list and one with a short list.
- One disease with no remedy yet, to test the empty state.
- Two diseases sharing a symptom (for example fever) so search returns several results.
- A search term that matches nothing.

Part 4: The link between frontend and backend [LINK]

This part is the shared agreement. The frontend builds against it now; the backend must return exactly this later. If a difference appears, fix the backend, not the structure.

4.1 Data structure

Item	Fields
Category	id, name, slug, description
Disease	id, category (name and slug), name, slug, short summary, description, symptoms, causes, risk factors, prevention, remedy (or an explicit no-remedy marker), created and updated dates
Remedy	id, overview, care options, precautions, urgent signs, source or reviewer, last reviewed date

- Lists are always lists, never missing; an empty list is allowed.
- A disease without a remedy carries an explicit no-remedy marker.
- Dates use one consistent format; slugs are lowercase, hyphen-separated and unique.

4.2 The services area

Action	Input	Result
Search diseases	Text, optional category	Matches plus total count
Get disease	Slug	Full disease with remedy, or not found
Get featured diseases	None	Short list for home page
Get categories	None	All categories with counts
Get category	Slug	Category and its diseases
Admin log in and out	Email, password	Success or failure message
Admin stats	None	Totals and recently updated items
Admin disease / remedy / category actions	Full data, or an id	Create, update, delete results

- **[FRONTEND]** For now, functions read sample data, wait briefly (to test loading states), and change data in memory only.
- **[BACKEND]** Later, each function maps to one real endpoint. Keep a function-to-endpoint list.
- All functions report success and failure the same way: a short message and, for forms, which field failed.
- All backend addresses live in one settings place on the frontend.

4.3 Search rules (mirrored by the backend)

- Trim text and ignore capital letters; partial words match ('mala' finds malaria).
- Search disease name, short summary, symptoms and category name.
- Rank name matches first, then symptom matches, then summary matches.
- Always return a total count; no matches returns an empty list with count zero, never an error.

4.4 Endpoints the frontend will eventually call

Action	Endpoint	Access
Search diseases	GET /api/diseases (text, optional category)	Public
One disease	GET /api/diseases/:slug	Public
Featured diseases	GET /api/diseases/featured	Public
All categories	GET /api/categories	Public
One category	GET /api/categories/:slug	Public
Log in / out	POST /api/admin/login, /logout	Admin
Dashboard totals	GET /api/admin/stats	Admin
List / create diseases	GET, POST /api/admin/diseases	Admin
Update / delete a disease	PUT, DELETE /api/admin/diseases/:id	Admin
Remedies	POST, PUT, DELETE /api/admin/remedies	Admin
Categories	POST, PUT, DELETE /api/admin/categories	Admin

4.5 Response rules

- Same structure as the frontend sample data.
- One consistent error format: status, short message and, for forms, which field failed.
- Standard meanings for success, created, bad input, not logged in, not allowed, not found, server problem.
- Lists are never null, only empty, so frontend empty states work.

Part 5: Backend plan [BACKEND]

Planning only. Nothing here is built until the frontend is finished, tested and handed over.

5.1 Goals

- Store disease, remedy and category information permanently.
- Serve it quickly to the public site, including search.
- Let only a logged-in administrator add, edit and delete records.
- Match the data structure the frontend already expects.

5.2 Technology

Piece	Choice
Language and framework	Node.js with Express (one language across the project)
Database	MySQL
Passwords	bcrypt hashing
Login method	Token or secure session cookie (decision recorded in B4)
Input checking	Server-side validation library
Secrets	Private settings file, never shared publicly
API testing	Postman or Thunder Client

5.3 How the system fits together

- The React frontend asks the backend for data through the services area.
- The backend checks the request, then reads or changes the database.
- The backend returns data in the agreed structure. The frontend never talks to the database directly.

5.4 Database design

Table	Fields
admins	id, name, email (unique), password (hashed), created date
categories	id, name (unique), slug (unique), description
diseases	id, category id, name, slug (unique), short summary, description, created and updated dates
symptoms, causes, risk_factors, preventions	id, disease id, text, display order (many per disease)
remedies	id, disease id (unique), overview, reviewed by or source, last reviewed date, updated date
remedy_care_options, remedy_precautions, remedy_urgent_signs	id, remedy id, text, display order

- Relationships: one category has many diseases; one disease has many list items and at most one remedy; one remedy has many care options, precautions and urgent signs.
- Deleting a disease also deletes its list items and remedy.

- Deleting a category is blocked while diseases use it, with a clear message.
- Slugs are unique and generated from names; the updated date changes on every edit.

5.5 Search design

- Mirror the rules in section 4.3 exactly.
- Suggestions use a lightweight version returning only names and slugs, capped at a handful.
- Always pass search text to the database the safe, parameterized way.
- Pagination is optional; add it if the database grows past roughly 30 to 50 diseases without changing the response structure.

5.6 Authentication and security

Area	Rule
Passwords	Stored only as hashes, with a minimum length rule
Login attempts	Repeated failures are slowed or paused
Admin routes	Every admin endpoint checks for a valid login before doing anything
Token or session	Expires after a set time, with a clear logout
Input checking	Every field validated on the server, even though the frontend also checks
Database safety	All queries are parameterized to prevent injection
Text safety	Admin text is cleaned so it cannot inject scripts into public pages
Cross-origin access	Accept requests only from your own frontend address
Secrets	Kept in a private settings file out of GitHub, never in logs or errors
HTTPS	Required once the site is live
Error messages	Never reveal database details or internal paths
Admin creation	First admin created directly by you; no public sign-up page

5.7 Content integrity rules

- A remedy cannot be saved without a source or reviewer and a last reviewed date.
- Only one remedy per disease, enforced by the database.
- The admin remedy form shows a reminder: do not include dosages.
- The dashboard shows review age so old content gets rechecked.

5.8 Hosting needs

Part	Needs
Frontend	Static hosting (Netlify, Vercel or similar)
Backend	Node hosting (Render, Railway or a VPS)
Database	Managed MySQL
Settings	Environment variables on the host, never in code
Backups	Regular database backups with a tested restore

Part 6: Frontend build plan [FRONTEND]

Eight milestones. Tags inside each milestone show where the backend or the link is involved.

Milestone 1: Setup and design foundations (3 to 4 sessions)

- **[FRONTEND]** Install Node.js, a code editor, Git and a GitHub account; create the project with Vite; add React Router and Tailwind.
- **[FRONTEND]** Create the planned folders; add palette, fonts and icon set to the Tailwind setup; build a style test page.
- **[FRONTEND]** Finalize the name, create logo files (full, icon-only, white, in SVG) and sketch wireframes for Home, Disease, Remedy and Admin Disease List.
- **[FRONTEND]** Habits: save to Git after each session, keep a 'later' list, write a short project notes file.

Done when: the project runs, the style test page matches the design plan, logo files and four wireframes exist, and work is on GitHub.

Milestone 2: Shared layout and data foundation (3 to 4 sessions)

- **[FRONTEND]** Build the public layout and admin layout; set up every route with placeholders and the Not Found route; add the placeholder login status that redirects admin pages.
- **[LINK]** Agree and write down the data structure (section 4.1) before writing data.
- **[FRONTEND]** Write the sample data: three categories and five diseases, with malaria first and the edge cases included.
- **[LINK]** Build the services area (section 4.2): one function per action, same result shapes the backend will return.
- **[BACKEND]** Note: sample data becomes the first database records, so write it as if it were real. Start the function-to-endpoint list.

Done when: both layouts work on phone and desktop, every route opens, the services area covers every action, and no page reads sample data directly.

Milestone 3: Home page and search (4 to 5 sessions)

- **[FRONTEND]** Build shared state components (button, tag, loading, empty state, error, breadcrumbs). Every page treats a result as loading, success, empty or error.
- **[FRONTEND]** Build the search bar (suggestions after 2 to 3 characters, brief wait after typing, keyboard support) and the home page sections, each handling its own loading and error state.
- **[FRONTEND]** Build the results page with category filter, count, no-results and error states; put the query and filter in the web address so it can be refreshed and shared.
- **[LINK]** Follow the agreed search rules (section 4.3).
- **[BACKEND]** Record needs: fast suggestions endpoint, a definition of 'featured' (a flag or most recently updated), a paging decision, and alternate names as a version 2 item.

Done when: searching 'malaria', 'fever', 'mala', an unknown word and an empty box all behave as planned.

Milestone 4: Disease and remedy pages (4 to 5 sessions)

- **[FRONTEND]** Build Info Section, Quick-jump menu, Warning Box (amber), Urgent Care Box (red), Source Note, call-to-action card and shared Disclaimer.
- **[FRONTEND]** Disease page: breadcrumbs, header, quick-jump, sections in order, call-to-action, urgent box, disclaimer; friendly not-found page; decide once how empty lists look.
- **[FRONTEND]** Remedy page: overview, care options, amber precautions, red urgent care box, source note; clear message when no remedy exists; optional 'may be out of date' note when review is over a year old.

- **[LINK]** One request loads a complete disease page, with the remedy inside it.
- **[BACKEND]** Record integrity rules: source and review date required, no dosages, review-age display for the dashboard.

Done when: all five diseases display correctly, including the long-list and no-remedy cases, and safety boxes are consistent.

Milestone 5: Category and About pages (2 to 3 sessions)

- **[FRONTEND]** All Categories and Single Category pages with all states, alphabetical sort and a friendly category-not-found page.
- **[FRONTEND]** About and disclaimer page using the shared disclaimer wording.
- **[LINK]** Counts are calculated from the data, never typed by hand.
- **[BACKEND]** Note: the categories endpoint must return disease counts; deleting a category in use is blocked.

Done when: the whole public site works end to end. This is the first showable checkpoint.

Milestone 6: Admin area (6 to 8 sessions)

Stage	Work	Part
A: Login and entry	Login screen with generic error message, show/hide password, redirects, logout, loading state	**[FRONTEND]**, real login later by **[BACKEND]**
B: Dashboard	Stat tiles, recently updated list, optional needs-attention list, quick actions	**[FRONTEND]**; stats from one efficient request (**[BACKEND]**)
C: Reusable pieces	Data table, table search, form field, repeatable list input, confirm-delete popup, notice bar, unsaved-changes warning	**[FRONTEND]**
D: Diseases	List, add/edit form with validation, slug preview, delete confirmation	**[FRONTEND]**; **[LINK]** whole disease sent in one piece, lists replaced as a whole; **[BACKEND]** repeats all validation
E: Remedies	List with review-age marker, form needing disease, source and review date, no-dosage reminder	**[FRONTEND]**; **[BACKEND]** refuses missing source or date, enforces one remedy per disease
F: Categories	Add, rename inline, delete with in-use block	**[FRONTEND]**; **[BACKEND]** enforces the block; slugs stay fixed after creation
G: Review	Run admin checks; test long, accented and symbol-heavy text; write the admin security requirements list	**[FRONTEND]**, **[LINK]**, **[BACKEND]**

Done when: every admin action works against the sample data with validation and messages, and all of it goes through the services area.

Milestone 7: Responsive, accessibility and polish (3 to 4 sessions)

- **[FRONTEND]** Responsive pass at phone, tablet and desktop widths, including rotation and 150% zoom.
- **[FRONTEND]** Accessibility pass: contrast, keyboard use, labels, heading order, text alternatives, descriptive links.
- **[FRONTEND]** Polish: favicon, page titles, share previews, spelling, consistent styling, remove the style test page, clear console warnings.
- **[BACKEND]** Note: page titles and share previews for each disease will come from real data later.

Milestone 8: Testing and handover (3 to 4 sessions)

- **[FRONTEND]** Run the full testing checklist (Part 8) in at least two browsers; fix and retest.
- **[FRONTEND]** Fresh-eyes test: ask one or two new people to find malaria symptoms, find when to seek care, list respiratory diseases, search for something missing, and (admin) add, edit and delete a disease.
- **[LINK] [BACKEND]** Prepare the handover package: final data structure, function-to-endpoint list, error format, search rules, admin security requirements, content rules, version 2 list, sample data file and frontend wishes list.
- **[FRONTEND]** Tag the final frontend version on GitHub.

Part 7: Backend build plan [BACKEND]

Starts only after Milestone 8. Stack: Node.js, Express and MySQL.

B1: Setup, database and sample data (3 to 4 sessions)

- **[BACKEND]** Create the backend folder beside the frontend; install Express, MySQL tools, bcrypt, token or session tool, validation library and a settings reader.
- **[BACKEND]** Create the private settings file and exclude it from GitHub before the first save; add a public example file with blank values.
- **[BACKEND]** Create the database, a limited database user and all tables with their rules (unique values, one remedy per disease, delete behavior, display order); keep creation steps repeatable.
- **[BACKEND]** Load the five sample diseases, three categories, and create the first admin directly with a hashed password; run alive and database checks.

B2: Public endpoints (4 to 5 sessions)

- **[BACKEND]** Build standard response and error shapes, central error handling, not-found handling, the cross-origin rule and safe request logging.
- **[BACKEND]** **[LINK]** One disease by slug (with remedy or no-remedy marker), all categories with counts, one category, featured diseases.
- **[BACKEND]** **[LINK]** Search and suggestions following section 4.3, with safe parameterized queries.
- **[BACKEND]** Test with a saved request collection: normal input, 'malaria', 'fever', 'mala', unknown word, empty text, wrong slug, no-remedy disease, special characters.
- **[LINK]** Compare every response field by field against the final data structure.

B3: Connect the public frontend pages (3 to 4 sessions)

- **[FRONTEND]** Put the backend address in one settings place with separate local and live values.
- **[FRONTEND]** **[LINK]** Swap one action at a time, testing the affected pages: categories, featured, disease, search and suggestions, single category.
- **[FRONTEND]** Check real network behavior: loading, stopped backend with retry, slow connection, quick repeated typing, refresh and back button.
- **[FRONTEND]** Re-run the public checks; pages should look identical to the sample-data version.

B4: Admin login and protection (3 to 4 sessions)

- **[BACKEND]** Login endpoint comparing the typed password with the stored hash, expiring logins, a generic failure message, limits on repeated failed attempts, and logout.
- **[BACKEND]** One protection check applied to the whole admin group of endpoints; record the decision between secure cookie and stored token.
- **[FRONTEND]** **[LINK]** Replace the placeholder login with the real one; on 'not logged in' return to the login screen with a clear message; check real status when admin pages open.
- **[BACKEND]** Test every admin endpoint with no login, wrong login and expired login; all must be rejected.

B5: Admin endpoints (5 to 6 sessions)

- **[BACKEND]** Write validation once from the handover list, with field-level errors, text cleaning, and server-generated slugs with duplicate handling.
- **[BACKEND]** **[LINK]** Disease list, create, update and delete; create and update save the disease and all lists as one all-or-nothing operation; update replaces lists as a whole; delete removes everything attached.

- **[BACKEND] [LINK]** Remedy actions with integrity rules: source and review date required, one remedy per disease, optionally no future review date.
- **[BACKEND] [LINK]** Category create and rename (slugs stay fixed) and delete blocked while in use; dashboard stats from efficient queries.
- **[BACKEND]** Extend the request collection: missing fields, over-long text, empty list items, duplicates, a failure halfway through, special characters.

B6: Connect the admin frontend pages (3 to 4 sessions)

- **[FRONTEND] [LINK]** Connect reading screens first (dashboard, disease list, remedy list, category list), then the disease and remedy forms, then categories.
- **[FRONTEND]** Show server field errors beside the right fields; confirm success notices, list refresh and the unsaved-changes warning.
- **[FRONTEND]** Test awkward cases: login expiring mid-form, backend stopped mid-save, double-clicking Save, duplicate names.
- **[LINK]** Confirm public pages show admin changes immediately; re-run the admin checks.

B7: Security review and full testing (3 to 4 sessions)

- **[BACKEND]** Verify the security table in section 5.6 item by item, and check installed tools for known security warnings.
- **[BACKEND]** Run the backend checklist (Part 8).
- **[FRONTEND] [BACKEND] [LINK]** Re-run the complete frontend checklist against the real backend, repeat the fresh-eyes test, test long text and slow connections.
- **[BACKEND]** Confirm a database backup can actually be restored.

B8: Deployment (2 to 3 sessions)

- **[BACKEND]** Choose hosts, create the live database, set new live secrets and a new strong admin password.
- **[BACKEND] [FRONTEND]** Deploy the backend, point the frontend at it, deploy the frontend, enable HTTPS, connect the domain, and confirm refreshing deep page addresses works.
- **[BACKEND]** Turn on regular automatic backups and basic monitoring; tag the live version on GitHub.

Part 8: Testing checklists

8.1 Frontend checklist [FRONTEND]

Global (every page)

- [ ] Navbar links work and highlight the current page; logo returns home
- [ ] Disclaimer banner and footer disclaimer appear; page titles match
- [ ] No broken images or icons; wrong addresses show Not Found
- [ ] Refresh keeps the user on the same page; back and forward behave sensibly

Page by page

- [ ] Home: search is the most noticeable element; Enter and button both work; empty search shows a gentle prompt; cards open the right pages; counts match the data
- [ ] Search results: query pre-filled; count correct; 'fever' returns several diseases; no-results message; filter works and clears; refreshed or shared address shows the same results
- [ ] Disease: sections in planned order; quick-jump links work; long and short lists both look clean; breadcrumbs correct
- [ ] Remedy: red box is the most striking element; source and date show; a disease with no remedy shows the empty state; back link works
- [ ] Categories: counts correct; single category lists only its diseases; sort works; empty category shows a message
- [ ] About: readable, clear sections, links work

Admin (sample data)

- [ ] Login shows errors for empty fields and a generic message for wrong details; admin pages cannot be reached without 'logging in'
- [ ] Dashboard totals match the data; table search works
- [ ] Add disease: required fields show errors; 'add another' and remove work for every list
- [ ] Edit opens with existing values; delete shows a confirmation popup and can be cancelled
- [ ] Remedy form requires a disease first; category add, rename and delete work, including the in-use block

Responsive, design and accessibility

- [ ] Phone: single column, no sideways scrolling, collapsing menu, easy tap targets; tablet: two columns; desktop: sticky quick-jump and sidebar filters
- [ ] Admin on phone: slide-out menu; tables scroll sideways inside their own area
- [ ] Palette respected; red and amber only on safety content; same fonts, buttons and spacing everywhere
- [ ] Strong contrast; warnings use icon and text; whole site usable by keyboard with visible focus; every field labeled; descriptive link text
- [ ] No errors in the browser console; favicon present; spelling checked

8.2 Content checks [CONTENT]

- [ ] Every disease and remedy has a source
- [ ] Lines marked 'verify' checked against NCDC or WHO
- [ ] No dosages anywhere; plain, consistent wording; disclaimer wording identical everywhere

8.3 Readiness for the backend [LINK]

- [ ] All data comes through the services area only

- [ ] Sample data matches the final structure exactly
- [ ] Every admin action already has a place in the services area
- [ ] Handover documents complete (structure, function-to-endpoint list, error format, search rules, security list, content rules, version 2 list, sample data file, frontend wishes list)

8.4 Backend checklist [BACKEND]

- [ ] Every public endpoint returns the right structure for all five diseases
- [ ] Search finds by name, symptom and category, and returns an empty list for no matches; a no-remedy disease returns correctly
- [ ] Admin endpoints reject no login and expired login; wrong password gives a generic error that does not reveal whether the email exists
- [ ] Creating with missing fields returns clear field errors; duplicates are handled gracefully
- [ ] Deleting a disease removes all related rows; deleting a category in use is blocked with a helpful message
- [ ] Very long text, symbols, apostrophes and accents save and display correctly
- [ ] Repeated failed logins are slowed
- [ ] The frontend works against the backend with no change except the services area

Part 9: Timeline, risks and open decisions

9.1 Timeline (about 4 sessions a week, 2 to 3 hours each)

Milestone	Focus	Part	Sessions
1	Setup and design foundations	[FRONTEND]	3 to 4
2	Shared layout and data foundation	[FRONTEND] [LINK]	3 to 4
3	Home page and search	[FRONTEND]	4 to 5
4	Disease and remedy pages	[FRONTEND]	4 to 5
5	Category and About pages	[FRONTEND]	2 to 3
6	Admin area	[FRONTEND]	6 to 8
7	Responsive, accessibility, polish	[FRONTEND]	3 to 4
8	Testing and handover	[FRONTEND] [LINK]	3 to 4
	Frontend phase: about 5 to 6 weeks		
B1	Setup, database, sample data	[BACKEND]	3 to 4
B2	Public endpoints	[BACKEND] [LINK]	4 to 5
B3	Connect public pages	[FRONTEND] [LINK]	3 to 4
B4	Admin login and protection	[BACKEND]	3 to 4
B5	Admin endpoints	[BACKEND] [LINK]	5 to 6
B6	Connect admin pages	[FRONTEND] [LINK]	3 to 4
B7	Security review and testing	[BACKEND]	3 to 4
B8	Deployment	[BACKEND]	2 to 3
	Backend phase: about 4 to 5 weeks. Total: about 9 to 11 weeks		

Minimum viable frontend if time is short: Milestones 1 to 5, plus a reduced Milestone 6 (login, dashboard, disease list and disease form only).

9.2 Risks

Risk	Part	What helps
Admin forms take longer than planned	[FRONTEND]	Finish the disease form completely, then reuse its pattern
Spending too long on design	[FRONTEND]	Follow the design plan; do not reopen decisions mid-build
Content writing slows the build	[CONTENT]	Finish malaria fully; keep the other four shorter but complete
Scope creep	[BOTH]	Keep a 'later' list; finish planned pages first
Frontend and backend data shapes drift apart	[LINK]	Treat the structure as fixed and fix the backend; compare field by field
Validation rules differ between form and server	[LINK]	Write rules once in the handover list and mirror them
Frontend security assumed to be enough	[BACKEND]	The frontend login is cosmetic; the server decides everything

Risk	Part	What helps
Half-saved records after a failure	[BACKEND]	Make every multi-table save all-or-nothing
Secrets leaked to GitHub	[BACKEND]	Exclude the private settings file first; use new secrets live
Admin endpoint left unprotected	[BACKEND]	Apply the login check to the whole admin group and test each endpoint without a login
Works locally, breaks live	[BOTH]	Test cross-origin settings, HTTPS and deep page addresses early in B8
Health information errors	[CONTENT]	Verify flagged lines and have a qualified person read the content

9.3 Open decisions

- [ ] Final site name and logo (working name: DiseaseWise) [FRONTEND]
- [ ] Start date for Milestone 1 [BOTH]
- [ ] Who will review the health content [CONTENT]
- [ ] Definition of 'featured' diseases: a flag or most recently updated [BACKEND]
- [ ] Login carried by secure cookie or stored token [BACKEND]

9.4 Version 2 ideas (not in the first build)

- [BACKEND] [FRONTEND] Alternate disease names (for example 'high blood pressure' finds hypertension).
- [BACKEND] Activity log, multiple admin roles, draft and published states, paging, automatic link checker.
- [BOTH] Feedback or report-a-problem form, printable pages, multiple languages, dark mode.